

Submission Worksheet

IT490-450-M2026 / mq-practice-individual

Submission Data

Course	IT490-450-M2026
Assignment	MQ Practice Individual
Student	Tristan D. (tad46)
Status	submitted
Worksheet Progress	100%
Potential Grade	10.00/10.00 (100.00%)
Received Grade	0.00/10.00 (0.00%)
Started	6/9/2026, 3:17:51 PM
Updated	6/9/2026, 6:16:05 PM

Grading Link

<https://courses.ethereallab.app/assignment/v3/IT490-450-M2026/mq-practice-individual/grading/tad46>

View Link

<https://courses.ethereallab.app/assignment/v3/IT490-450-M2026/mq-practice-individual/tad46>

Instructions

1. Read all instructions and requirements first
2. Use any VM creation tool that gives you root access and persistent storage
 - VirtualBox, Multipass, cloud (Amazon, Google, Azure, etc) (Docker won't be an option here)
 - Create a hostname relevant to the assignment (i.e., test-individual)
 - Create a user of your ucid with a password, ensure relevant permissions
 - Hardware: 1GB Memory, 10GB Hard Drive
 - Install a server version of linux (i.e., Ubuntu Server 24.04)
 - **Hint:** You may want to get a base install working and use that as a cloning point for quicker destroy/create cycles
3. Get the sample code from the `main` branch at <https://github.com/MattToegel/IT490-2026>
4. Connect to one VM with two separate ssh connections
 - Run the `RabbitMQServerSample.php` file successfully in one instance
 - Run the `RabbitMQClientSample.php` file successfully in another instance
 - Proper data should be sent/received
5. Create a `setup.sh` script that automates the installation/setup logic to get this sample working (steps needed after the installation and login of Ubuntu)
6. Create a branch in the format of `module02-<ucid>-individual-practice` in your team's repository (provided by the instructor)
7. Upload the content of the `setup.sh` script to it and the `composor.json`
8. Fill in the below requirements
9. Submit and Export once done
10. Upload the PDF to the branch you created above in your group GitHub repo for the class
11. No need to merge (Pull Request) this branch
12. Upload the PDF to Canvas
13. You may delete any test VMs created for this assignment

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Submission Progress

100%

Section #1: (7 pts.) Example Solution

100%

Task #1 (3.5 pts.) – Demonstrate successful example

100%

Details

- Run the echo example and capture both the server and client output
- Run the ping/pong example and capture both the server and client output
- UCID (user) must be present
- Clearly caption each image



echo example with both the server and client output



ping/pong example with both the server and client output

Task #2 (3.5 pts.) – Setup Script

100%

Details

- This script must do all of the install/setup steps in order to immediately test the example code
 - Installing any tools needed (i.e., RabbitMQ server, etc)
 - Downloading the example code from my GitHub (including unzipping if necessary for your workflow)
 - Any PHP/Composer processes
 - The script should be able to be executed multiple times without having or causing an error
- Setup the composer.json to include a script section to be able to invoke this script via the `composer` command

Combo #1

Part 1: Images 100%

Details

- Show the content of the `setup.sh` script you created to automate the installation and configuration steps that lead up to a working example.
 - Include comments noting what each line is doing
 - Comments should contain your ucid
- Show the raw output of the script being freshly executed successfully (may be best to pipe it to a file and capture the content)
- Include a screenshot of the modified `composer.json`
- Clearly caption each image



```
#!/bin/bash
# Setup script for the assignment
# This script will install the necessary dependencies and
# set up the environment for the assignment.

# Install the necessary dependencies
sudo apt-get update
sudo apt-get install -y php php-cli php-fpm php-mysql php-gd php-imagick php-redis php-symfony/php-composer

# Set up the environment
export COMPOSER_HOME=/home/$USER/.composer
export PATH=$PATH:/home/$USER/.composer/vendor/bin
```

Created setup.sh script

```
#!/bin/bash
# Script to run the assignment
# This script will run the assignment and
# save the output to a file.

# Run the assignment
php /home/$USER/assignment.php > /home/$USER/assignment_output.txt
```

script execution piped to file

```
{
  "name": "matt-toegel/it490-m2026-autopilot-engineers",
  "description": "Personal IT490 Autopilot response engine sample.",
  "type": "project",
  "require": {
    "php": ">=7.4",
    "symfony/php-composer": "1.*"
  },
  "scripts": {
    "start": "php /home/$USER/assignment.php"
  }
}
```

edited composer.json to run script

Part 2: URLs 100%

Details

- Include a link to the `setup.sh` script from your personal branch on the group repository
- Include a direct link to your modified `composer.json`

URL #1

<https://github.com/MattToegel/it490-m2026-autopilot-engineers/blob/module02-tad46-individual-practice/docs/module02-tad46/setup.sh> 🍏

URL #2

<https://github.com/MattToegel/it490-m2026-autopilot-engineers/blob/module02-tad46-individual-practice/docs/module02-tad46/composer.json> 🍏

Section #2: (3 pts.) Reflection

100%

Task #1 (1 pt.) – What was the easiest part of this assignment

100%

Details

- At least a few thoughtful sentences

Response

The easiest part of this assignment was that once everything was setup, all I had to do was run a few commands. After making sure everything was installed, I just had to run the php commands for rabbitmq client and server examples and run the script.

Supports **bold**, *italic*, [links](url), `code`, etc.

250 chars

Task #2 (1 pt.) – What was the hardest part of this assignment

100%

Details

100%

Details

- At least a few thoughtful sentences

Response

The hardest part of the assignment was making sure everything was installed before hand. I also had to make the rabbitmq server was set up correctly. Additionally, I have a captcha issue on my Github account and I don't have a password tied to it but my email, so I had to generate a PAT, **Personal Access Token**, on Github so that I could push the files to the github branch on my vm. I also had to look into "if" statements of scripts for the assignment so that it ran well and didn't flood the machine with a bunch of errors or extra unnecessary information.

Supports **bold**, *italic*, [links](url), ``code``, etc.

565 chars

Task #3 (1 pt.) – What did you learn during this assignment

100%

Details

- At least a few thoughtful sentences

Response

This assignment gave me hands-on experience setting up RabbitMQ on a Linux VM and using a basic client/server messaging structure. I was able to learn how to effectively write a script and it showed how automation can make environment setup repeatable and reliable. I was also able to test how to SSH into a vm with Tailscale, which is important for our team project. Working with GitHub branches also built on my skills with basic commands used in real software development.

Supports **bold**, *italic*, [links](url), ``code``, etc.

476 chars

End of Assignment — make sure you click **Submit**.